

Ayomide Oshilaja

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PROFESSIONAL SUMMARY

Computer Science junior at Arizona State University pursuing a Student Researcher position, with a focus on AI/ML systems, applied language model integrations, and software engineering for data-intensive systems. Hands on experience building and evaluating retrieval augmented generation pipelines, streaming LLM architectures, and full stack systems using TypeScript, React, Next.js, Python, and the Anthropic Claude API. Combines production engineering depth with a growing research practice, designing experiments, comparing system approaches, and documenting tradeoffs.

EDUCATION

Arizona State University, Ira A. Fulton Schools of Engineering

Bachelor of Science in Computer Science

Expected 12/27

Tempe, AZ

- Relevant Coursework: Operating Systems (CSE 330), Programming Languages (CSE 340), Theory of Computation (CSE 355), Information Assurance and Cybersecurity (CSE 365), Data Structures, Object Oriented Programming.
- Honors: WICS Top 3 Programming Competition; WICS Mentorship (Spring 2026).

PROFESSIONAL EXPERIENCE

Software Engineer Intern

JUSTPLATYPOST LLC

March 2026 to Present

Remote

- Engineered the AI/ML foundation and core internal tools of the Tukd platform in TypeScript, building a streaming Anthropic Claude SDK wrapper with prompt caching and tool use loop orchestration that powers 100% of downstream code generation.
- Designed and evaluated Server-Sent Events (SSE) streaming API routes for real-time text deltas, tool events, and file update notifications, comparing latency tradeoffs across caching strategies to achieve sub-200 ms first-token latency.
- Redesigned the product UI/UX in Figma and shipped a cross-platform design system of 20+ reusable Tailwind CSS and React components spanning the Next.js landing page and React Native / Expo mobile preview.
- Practiced full Agile/Scrum workflow, opening 10+ pull requests, addressing code review feedback, and resolving deployment issues through twice-weekly standups.

Code Coach

TheCoderSchool

April 2026 to Present

Tempe, AZ

- Teach 1 on 1 and small group coding lessons to K-12 students, translating programming concepts such as loops, functions, and object-oriented design into age appropriate, project-based instruction.
- Design individualized learning plans and debug student projects in real time across Python, Scratch, and JavaScript, adapting explanations to each student's pace and learning style.

RESEARCH AND TECHNICAL PROJECTS

Polarix - MedPolicy: Retrieval Augmented Generation for Healthcare Policy Search

April 2026

Next.js, FastAPI, Claude API, pgvector (Hackathon)

- Investigated retrieval quality and answer synthesis for a RAG pipeline over 100+ healthcare policy documents, using the Claude API for synthesis and pgvector for embedding based retrieval.
- Evaluated chunking strategy and retrieval accuracy tradeoffs under a 48-hour build constraint and deployed the resulting Next.js frontend and FastAPI backend for a live hackathon demo in partnership with Anton RX.

Pulse-Play: BPM-Driven Music Discovery Web App

Feb 2026 to Apr 2026

Next.js, TypeScript, Tailwind, Supabase, Arduino, Web Serial API

- Prototyped a real-time recommendation system mapping heart rate BPM, captured from an Arduino pulse sensor via the Web Serial API, to 5 genre-zoned playlists, and shipped a publicly accessible full stack product with a Supabase-backed dashboard.

SafeTrack: GPS Personal Safety Desktop App

Aug 2025 to Present

Python, PyQt6, SQLite, Mesh Networking (ASU EPICS)

- Lead a four-person engineering team defining architecture and applying MVC and observer design patterns for an offline-capable, mesh-networked safety application.

TECHNICAL SKILLS

Languages and Frontend: Python, Java, JavaScript, TypeScript, SQL, HTML5, CSS3, Bash; React, Next.js, React Native, Expo, Tailwind CSS, Figma, UI/UX design.

AI/ML and Backend: Anthropic Claude API (streaming, prompt caching, tool use), Google Gemini API, artificial intelligence integrations, agentic AI, RAG, pgvector, Server-Sent Events (SSE), REST APIs, Node.js, Flask, FastAPI, PyQt6.

Cloud, Tools and Methodologies: Vercel, Render, Firebase, Supabase, Stripe, EAS Build, Git, GitHub, GitHub Actions, VS Code, Agile/Scrum, code review, CI/CD, test driven development, Linux/Unix.